

Daily Geometry Drill #3

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions (4/3/2/1 tangents) · Radical Axis by subtraction · Common Chords · Orthogonal Circles ($d^2 = r_1^2 + r_2^2$).

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Compare the centre distance d with $r_1 + r_2$ and $|r_1 - r_2|$, or subtract the equations to read the radical axis.

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.
2. The circles with centre $(0,0)$ radius 5 and centre $(13,0)$ radius 3 are disjoint. Write down the number of common tangents of the two circles.
3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y - 11 = 0$ and $x^2 + y^2 - 2x - 4y + 1 = 0$.
4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).
5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.
6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.
7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .
8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .
9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?
10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Section B Manipulation Drills

(≤50 s each)

Complete the square, locate the centres, then classify the two-circle interaction or compute the radical axis.

1. How many common tangents do the circles $x^2 + y^2 - 2x - 4y - 20 = 0$ and $x^2 + y^2 - 6x - 2y + 6 = 0$ have?
A) 0
B) 1
C) 2
D) 3
E) 4
2. The circle $x^2 + y^2 + ax + by + c = 0$ passes through the origin, and its radical axis with the circle $x^2 + y^2 + 6x - 4y + 10 = 0$ is the line $2x - y + 5 = 0$. What is $a + b$?
A) 3
B) 5
C) -3
D) 0
3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .
A) -9
B) 9
C) -5
D) 14
4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.
5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?
A) $x = 3$
B) $x = 6$
C) $y = 3$
D) $y = 0$
6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?
A) 0
B) 1
C) 2
D) 3
E) 4

7. A circle with centre $(7, 1)$ cuts the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ at right angles. What is its radius?
- A) $3\sqrt{5}$
B) $3\sqrt{5} - 3$
C) 3
D) 6
8. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 12x - 6y + 25 = 0$ meet at P and Q . Find the midpoint of PQ .
9. A circle with its centre in the first quadrant touches both coordinate axes and the line $3x + 4y = 12$. What is the sum of the possible radii?
- A) 1
B) 6
C) 7
D) 5
10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Section C Substitution & Structure(≤ 75 s each)

Hard, non-routine: two or three circle tools at once — subtract for the radical axis, then turn that line into a chord or a tangency count. No brute intersection.

1. Two circles have the same radius. Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which is the equation of the line PQ ? (after TMUA 2021 Paper 1 Q1)
- A) $5x + 3y = 4$
B) $3x - 5y = 4$
C) $5x - 3y = 4$
D) $5x - 3y = 1$
2. The circles $C_1 : (x - 1)^2 + y^2 = 4$ and $C_2 : (x - 5)^2 + (y - 3)^2 = 9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have? (after TMUA 2019 Paper 1 Q7)
- A) 0
B) 1
C) 2
D) 3
E) 4
3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q . What is the length PQ ?

- A) 12
B) 16
C) 24
D) 26
4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$? (*adapted from JZMaths Mock 1 Q5*)
A) 14
B) 17
C) 22
D) 34
5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 16x - 8y - 1 = 0$ meet at P and Q . What is the radius of the circle through P , Q and the point $(0, 1)$?
A) 9
B) $2\sqrt{13}$
C) $4\sqrt{5}$
D) 13
6. A circle has equation $x^2 + ax + y^2 + by + c = 0$ where $a, b, c \neq 0$. Which condition is necessary and sufficient for the circle to be tangent to the y -axis? (*after TMUA 2021 Paper 2 Q15*)
A) $a^2 = 4c$
B) $b^2 = 4c$
C) $a^2 + b^2 = 4c$
D) $c = 0$
7. The equal circles $C_1 : (x + 2)^2 + (y - 1)^2 = 3$ and $C_2 : (x - 4)^2 + (y - 1)^2 = 3$ have centres 6 apart. A transverse common tangent with positive gradient makes an acute angle θ with the x -axis. What is $\tan \theta$? (*adapted from TMUA 2020 Paper 1 Q16*)
A) $\frac{1}{2}$
B) $\frac{\sqrt{2}}{2}$
C) $\sqrt{2}$
D) $\frac{\sqrt{3}}{3}$
8. The three circles $C_1 : x^2 + y^2 + 4x - 1 = 0$, $C_2 : x^2 + y^2 - 8y + 11 = 0$, $C_3 : x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes concurrent at R . What are the coordinates of R ? (*adapted from Vantage Mock Paper 2 Q10 / BMO1 2017 Q4*)
A) $(1, 1)$

- B) $(1, -1)$
- C) $(3, 1)$
- D) $(1, 2)$

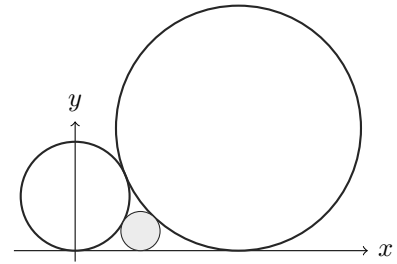
Section D Challenge Ramp $(\leq 90$ s each)

Hardest 5: combine radical axes, orthogonality and chord loops end-to-end. The numbers still cancel, but the route is hidden.

1. The circles $x^2 + y^2 - 4x - 2y - 4 = 0$ and $x^2 + y^2 - 20x - 14y + 124 = 0$ have centres O_1 and O_2 . A common external tangent touches them at P and Q respectively. What is the area of the quadrilateral O_1PQO_2 ?
 - A) 24
 - B) $16\sqrt{6}$
 - C) 40
 - D) $8\sqrt{6}$
2. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (*after TMUA 2019 Paper 1 Q6*)
 - A) 8
 - B) 16
 - C) 26
 - D) 42
3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres? (*adapted from BMO1 2017 Q4*)
 - A) $12\sqrt{5}$
 - B) 12
 - C) 84
 - D) 30
4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?
 - A) $\frac{12}{5}$
 - B) $\frac{24}{5}$
 - C) 6
 - D) $\frac{48}{5}$

5. The circles $x^2 + y^2 - 8y = 0$ and $x^2 + y^2 - 24x - 18y + 144 = 0$ both touch the x -axis and touch each other. A third circle lies in the gap between them and the x -axis, and touches both circles and the x -axis. What is its radius?

- A) $\frac{6}{5}$
 B) $\frac{36}{13}$
 C) $\frac{169}{100}$
 D) $\frac{36}{25}$



“Two circles are decoded by one subtraction: locate each centre, subtract the equations, then compare d with $r_1 + r_2$ and $|r_1 - r_2|$.”

Found a mistake or want to contribute a sheet?
 Visit the open-source project at github.com/The-CerealDev/speed-maths